

Assignment 2: Randomized Algorithms

Randomized Quicksort | Randomized Select | Karger's Min Cut

Assigned	Tuesday, September 29, 2026
Due	Tuesday, October 13, 2026, 6:00 PM (start of Week 5 class)
Submit	Source code + one short PDF report in Brightspace
Readings	Levitin §5.2; CLRS Chapters 5, 7, 9; class notes for Karger's Min Cut

Purpose: Implement and analyze randomized algorithms, measure how random choices affect performance, and explain the difference between Randomized Quicksort and Randomized Select.

Grading Overview

Part	Work	Points
A	Implement Randomized Quicksort, Randomized Select, and Karger's Min Cut	50
B	Experimental analysis of Quicksort and Karger's algorithm	30
C	Hand analysis and probability reasoning	20

Part A - Implement (50 points)

Use one consistent partition routine for Randomized Quicksort and Randomized Select. Your program must allow repeated experiments with a controllable random seed.

A1. Randomized Quicksort (15 points)

- Choose the pivot uniformly at random from the current subarray.
- Partition the subarray around the pivot, then recursively process both sides.
- Instrument the implementation to count element-to-pivot comparisons. This count will be used in Part B.
- Verify correctness on at least 30 randomly generated arrays by comparing your output with the language's built-in sorting result.

A2. Randomized Select (15 points)

- Implement Randomized Select to return the k -th smallest element, where k is 1-based.
- Use a random pivot and the same partitioning idea as Randomized Quicksort.
- After partitioning, recurse only into the side that contains the requested rank. Do not recursively process both sides.
- Verify correctness on at least 30 random arrays and random valid k values using `sorted(A)[k-1]` as the check.

Important distinction: Randomized Quicksort partitions and continues on both sides because it must sort the entire array. Randomized Select partitions but continues on only one side because it needs only the k -th smallest element.

A3. Karger's Randomized Min-Cut (20 points)

- Represent an undirected multigraph. During contraction, parallel edges must be retained and self-loops must be removed.
- At each contraction step, choose one current edge uniformly at random from the current edge multiset.
- Contract its endpoints into one supernode and continue until exactly two supernodes remain.

- Return the number of crossing edges between the final two supernodes as the cut size.
- Write a wrapper that runs the contraction algorithm independently r times and returns the smallest cut found.

Test graph for Karger (used again in Part B): two K_4 subgraphs joined by two bridge edges. The true minimum cut is 2.

Subgraph / bridges	Edges
Left K_4 on {A,B,C,D}	AB, AC, AD, BC, BD, CD
Right K_4 on {E,F,G,H} + bridges	EF, EG, EH, FG, FH, GH, DE, CF

Part B - Measure and Analyze (30 points)

B1. Randomized Quicksort vs. a deterministic pivot rule (15 points)

Create sorted arrays of distinct integers with the following sizes:

$n = 100$	$n = 250$	$n = 500$	$n = 750$
sorted input	sorted input	sorted input	sorted input

- Deterministic version: always choose the first element of the current subarray as pivot. Run once for each n and record the comparison count.
- Randomized version: on the same sorted input, run 30 independent trials for each n . Report the mean comparison count and the minimum and maximum observed.
- Create one plot with n on the x-axis and comparisons on the y-axis. Plot the deterministic count and the randomized mean.
- In 3-5 sentences, explain why the deterministic curve is quadratic on sorted input while the randomized curve is expected to grow on the order of $n \log n$.

B2. Karger's success probability in practice (15 points)

- Use the 8-vertex graph from Part A3. Its minimum cut is 2.
- Run 500 independent single trials of Karger's contraction algorithm.
- Report how many trials returned cut size 2 and the empirical success rate.
- Also report the frequency of every cut size you observed in a small table.
- Compare your empirical single-run success rate with the theoretical lower bound $2/[n(n-1)]$ for $n = 8$. The empirical rate may be higher; the bound is a worst-case guarantee.

Part C - Solve and Explain by Hand (20 points)

C1. Quicksort vs. Randomized Select on the same partition (10 points)

Use the array: [9, 4, 12, 3, 7, 15, 1, 10]. For this hand exercise, the pivot choices are supplied so everyone analyzes the same execution.

1. First pivot = 7. Partition the values into < 7 , pivot 7, and > 7 . Internal order inside each side does not matter.
2. For Randomized Quicksort, state the two recursive subproblems after this partition.
3. For Randomized Select, find the 5th smallest element. After pivot 7, identify which side is searched and convert k to the correct local rank.
4. Continue Randomized Select using pivot 12, then pivot 10. Show the rank calculation at every step and report the final answer.

C2. Karger's contraction and probability (10 points)

5. On K_4 with vertices {A,B,C,D}, contract edge AB first, then contract the supernode {AB} with C. Draw or list the remaining multigraph. How many parallel edges remain between {ABC} and D? What cut size is returned?
6. For a graph with $n = 8$ vertices, use Karger's lower bound to compute the guaranteed probability that one run finds a fixed minimum cut.
7. If 100 independent runs are performed, compute the lower bound on the probability that at least one run finds that minimum cut: $1 - (1 - 2/[n(n-1)])^{100}$. Give the result to at least three decimal places.
8. In one or two sentences, explain why parallel edges must be kept during contractions.

Check value: For $n = 8$, the one-run lower bound is $1/28 \approx 0.0357$. For 100 independent runs, the lower bound on at least one success is approximately 0.974. Show your own calculation in the submitted report.

Submission Requirements

- One source-code submission containing all three implementations and the experiment code.
- One short PDF report containing: verification results, the Part B plot and tables, your written analysis, and all Part C work.
- Label axes and tables clearly. Include the random seed(s) or describe how seeds were generated so the experiment can be reproduced.
- Your program output must make it possible to verify the reported results. Do not submit screenshots of code in place of source files.

What I will grade

Criterion	What counts	Points
Algorithm correctness	Correct random-pivot behavior, Select rank logic, and correct contraction handling	35
Implementation quality	Readable code, sensible decomposition, reproducible tests	15
Experimental method	Correct comparison counts, independent trials, correct Karger test graph and reporting	20
Analysis	Accurate interpretation of expected behavior and empirical results	10
Hand work	Correct partition/rank trace, contraction trace, and probability calculations	20

Course Policies for This Assignment

AI policy. You may use generative AI tools to explain concepts, explore alternative approaches, debug error messages, and review your own code or proofs. You may not use AI to generate assignment solutions. Any AI-assisted portion of submitted work must be labeled in a comment or footnote stating the tool and how it was used.

Late work. Late assignments are penalized 10% for each day delayed. In exceptional circumstances, request an extension before the due date.

Individual work. You may discuss course concepts with classmates, but the submitted code, analysis, plots, calculations, and explanations must be your own work unless the instructor explicitly states otherwise.

Before You Submit

- Randomized Quicksort uses a random pivot and sorts both partitions.
- Randomized Select searches only the partition containing the requested rank.
- Karger chooses uniformly from the current edge multiset, keeps parallel edges, and removes self-loops.
- All correctness tests pass.
- The Part B plot has labels and a legend.
- Part C calculations are shown, not just final answers.
- Any permitted AI assistance is disclosed.

Course alignment: Assignment 2 is assigned in Week 4 (Sep 29), after Randomized Quicksort, Randomized Select, and Karger's Min Cut. October 6 is a Wellbeing Day with no class, and the next class meeting is October 13.